TECHNICAL HANDOFF

# CARLA Nav2 AVL

Perception, costmap, Navigation2, and operator startup guide for the AVL IGVC ground robot

| Document | Value |
|---|---|
| Repository | arassal/carla-nav2-avl |
| Reviewed branch | main |
| Primary source snapshot | carla-nav2-avl 2684f876d503ce222289d5a3f2bb824492ea9fda (2684f87), committed July 29, 2026 |
| Vehicle source snapshot | IGVC_ROS2 d40e18aa480f76673b46a379ab7cb4618f279d75 (d40e18a), committed June 1, 2026 |
| Review date | August 5, 2026 |
| Intended recipient | Developer or technical lead assuming ownership |
| Document status | Expanded draft for review - includes operator runbook; not an operational sign-off |

Purpose and limitation

This document translates the repository into an ownership handoff. It distinguishes active implementation from legacy/stub code and flags documentation drift. It does not assert the current physical state of the robot, team availability, credentials, or safety approval. Those items require confirmation by the project owner.

# 0. Contents and first-day path



## Recommended first-day sequence

- Read sections 1, 4, 5, and 11 before touching code or the robot.
- Confirm the current vehicle state with the project lead and inspect the deployed commit on dinosaur.
- Run the offline perception tests in Ubuntu 22.04 / ROS 2 Humble, then build only perception_costmap.
- Bring up sensors and perception with Nav2 disarmed. Verify topics, TF, and visual outputs before any motion.
- Only arm Navigation2 with a human on the E-stop and a documented field-test plan.

> Highest-value fact
> The maintained core is perception_costmap plus driving_seg. Most other ROS packages and the root scripts directory are legacy or stubs. Start from the maintained package, not from the apparent breadth of the repository.

# 1. Executive summary

The project provides a ROS 2 perception-to-navigation pipeline for a slow skid-steer IGVC robot operating on grass between painted white lines and traffic cones. CARLA supports simulation work on x86, while the physical vehicle ('dinosaur') uses an NVIDIA Jetson AGX Orin with three ZED X cameras, a VLP-16 lidar, an Xsens IMU/GNSS, and a Teensy/SparkMAX drive interface. CARLA does not run on the Jetson; the intended portability boundary is the sensor source and actuator sink.

| SENSOR SOURCES | PERCEPTION CORE | NAVIGATION + CONTROL |
|---|---|---|
| 3x ZED X RGB cameras<br>VLP-16 lidar (currently disabled in active config)<br>TF and EKF from external avros bring-up | Segmentation + object detection<br>Per-camera IPM into fused BEV<br>Temporal filter + cost shaping<br>/perception/costmap | costmap_to_cloud bridge<br>Nav2 ObstacleLayer + inflation<br>Planner + regulated pure pursuit<br>/cmd_vel (isolated by default) |

Operationally, the active real-car path is camera-led: the current vehicle configuration enables HSV road segmentation, YOLO obstacles, cones, and white-line handling, but sets use_lidar: false. The perception grid is converted to a PointCloud2 stream for Nav2's ObstacleLayer. Navigation output is remapped away from the actuator topic unless NAV2_ARMED=1 is explicitly set.

## Status at handoff

| Area | Assessment | Evidence / implication |
|---|---|---|
| Perception core | Implemented and documented as maintained | ROS-free geometry/segmentation/occupancy modules, ROS node, configs, tools, and offline tests are present. |
| Real-car operations | Substantial operational glue exists | Boot service, tmux bring-up, camera recovery, dashboards, and Nav2 launch scripts are checked in; many are host-specific. |
| Performance | Acceptance not yet demonstrated in current snapshot | Last recorded TwinLite CUDA run was about 4-5 Hz against an 8 Hz target; current default changed to HSV and needs a fresh benchmark. |
| Geometry | Values exist; field truth still needs confirmation | Current dinosaur YAML includes camera mount/calibration values dated July 9, while multiple docs still call calibration open. |
| Navigation safety | Safe default, field validation required | Nav2 is disarmed by default; road keeping depends on a cost threshold contract between perception and cloud conversion. |
| Documentation | Useful but internally inconsistent | Older design/status documents describe obsolete StaticLayer and phase gaps. The active config and launch files should win. |

# 2. Scope and system boundaries

## In scope

- Three-camera road, white-line, person/vehicle, and cone perception.
- Bird's-eye-view fusion into a robot-centric OccupancyGrid.
- Cost shaping, temporal filtering, and per-class safety halos.
- Conversion of the perception grid to obstacle points consumed by Navigation2.
- Local/global rolling Nav2 costmaps, path planning, and regulated pure pursuit control.
- CARLA feed/test helpers, Jetson deployment guidance, boot automation, visualization, and recovery scripts.
- Standalone multi-model driving_seg research/tooling for richer semantic highlighting and course-class training.

## External or intentionally excluded

| Boundary | Owner / source | Handoff consequence |
|---|---|---|
| Physical sensor drivers, TF, and EKF | Separate IGVC_ROS2 / avros bring-up repository | This repository cannot produce live data without that stack. Confirm compatible branch and topic names. |
| Physical motor actuation | Existing actuator/CAN stack on the robot | Do not run duplicate actuator nodes. Navigation output is intentionally isolated until armed. |
| TensorRT engines | Built on the target Jetson | The repository ships model weights but engines are GPU/CUDA/TensorRT-specific and must be regenerated. |
| Credentials and network access | Project owner | No passwords or secrets belong in this handoff. Obtain robot SSH/Tailscale access separately. |
| Competition safety approval | Team safety lead | Software presence is not authorization to drive. E-stop and supervised-test procedures must be confirmed. |

## Authoritative-source order

- For runtime behavior: active launch files and active YAML configuration.
- For algorithm behavior: perception_costmap source and its tests.
- For actual deployment: inspect the files and commit running on dinosaur, then compare with main.
- For intent/history: README, DESIGN.md, DEPLOY.md, REPRODUCE.md, and FINALIZE.md, with dates considered.

# 3. Architecture and runtime data flow

## Perception flow

- Each configured camera subscribes to rectified RGB and CameraInfo using sensor-data QoS.
- A selectable segmenter produces the road mask. The active dinosaur config defaults to HSV; TwinLiteNet is available but requires external model code and a valid engine.
- YOLO produces class-aware camera obstacle masks. Cone detection is separately configured; white-line pixels are treated as non-drivable boundaries.
- Each camera mask is projected through inverse perspective mapping into a shared robot-centric bird's-eye-view grid. Camera contributions are fused.
- A temporal filter suppresses obstacle flicker. Occupancy shaping assigns free, off-road, unknown, lethal, and inflated per-class costs.
- The node publishes /perception/costmap, /perception/known, and /perception/obstacle_points.

## Active Nav2 flow

The current real_nav2_params.yaml deliberately omits StaticLayer. A bridge raycasts the robot-relative perception grid into /perception/costmap_cloud. Both rolling costmaps consume this cloud through ObstacleLayer, then apply InflationLayer. Navfn produces the path and RegulatedPurePursuitController follows it.

> Do not revive the obsolete StaticLayer design casually
> Older DESIGN.md text describes feeding /perception/costmap to StaticLayer. The active Nav2 config explains why that approach was abandoned: a rotating robot-relative grid cannot be represented correctly by the axis-aligned StaticLayer. Treat real_nav2_params.yaml and real_nav2_launch.py as the current design.

### Key topics and frames

| Interface | Type / frame | Direction and role |
|---|---|---|
| /zed_front/.../image and camera_info | sensor_msgs/Image, CameraInfo | Input; analogous left/right topics are defined in dinosaur YAML. |
| /velodyne_points | sensor_msgs/PointCloud2 | Configured lidar input, but ignored while use_lidar is false. |
| /perception/costmap | nav_msgs/OccupancyGrid; base_link | Primary perception result, 20 m x 20 m at 0.1 m/cell in the vehicle config. |
| /perception/costmap_cloud | sensor_msgs/PointCloud2 | Bridge output and active Nav2 obstacle source. |
| /odometry/filtered | nav_msgs/Odometry; odom/base_link | Vehicle localization used by run helpers and Nav2. |
| /nav2/cmd_vel | geometry_msgs/Twist | Default disarmed Nav2 output. |
| /cmd_vel | geometry_msgs/Twist | Real actuator command only when explicitly armed. |
| map -> odom -> base_link -> sensors | TF tree | Perception publishes in base_link; active Nav2 global frame is odom because no map anchor exists. |

# 4. Repository map and ownership

| Path | Status | Use |
|---|---|---|
| ros2_ws/src/perception_costmap/ | Maintained core | Perception node, ROS-free algorithm modules, launch, configs, tests, deployment tools. |
| driving_seg/ | Active research/tooling | Multi-model semantic highlighting and course-class training for cones and white lines. |
| models/ | Required artifacts | Committed YOLO and TwinLite weights. Target-specific TensorRT engines are not committed. |
| ros2_ws/src/carla_msgs/ | Simulation support | CARLA vehicle-control message definition. |
| ros2_ws/src/world_setup/ | Legacy/stub | CARLA bridge/world setup; not part of the maintained real-car path. |
| collision_guard, controller, route_planner, sdc_bringup, sdc_common | Legacy/stub | Do not extend without explicit review. Package presence does not imply readiness. |
| scripts/ | Legacy CARLA helpers | Historical x86 simulation launch and validation scripts; not the primary real-car runbook. |
| perception/ | Prototype | Original prototype scripts retained for attribution/history. |
| docs/plans, FINALIZE.md | Planning/history | Useful intent and acceptance ideas; may lag implementation. |

### Core perception modules

| Module | Responsibility |
|---|---|
| costmap_node.py | ROS wiring, parameters, subscriptions, staleness handling, multi-camera fusion, publishers. |
| segmentation.py | HSV and TwinLite segmentation backends, including TensorRT adapter paths. |
| obstacles.py | YOLO class-aware obstacle masks, cone detector, and lidar filtering helpers. |
| bev.py | IPM/homography math and camera-to-grid projection. |
| temporal.py | Per-cell confidence filter for transient obstacles. |
| occupancy.py | Cost-array construction, off-road shaping, unknown infill, and per-class danger zones. |
| deploy/costmap_to_cloud.py | Current perception-to-Nav2 bridge. |
| deploy/real_nav2_launch.py | Starts bridge and Nav2 servers; applies the safe command-topic remap. |

### Team metadata in repository

The repository lists Alexander (@arassal) as project lead, with jchy05, AdamCastillo07, and Adrian (@Ad-Tap) as contributors/mentees. Confirm current roles, availability, and operational ownership before treating these as escalation contacts.

# 5. Current state and evidence

## Snapshot

| Item | Observed in reviewed main snapshot |
|---|---|
| Commit | 2684f876d503ce222289d5a3f2bb824492ea9fda - Add .mailmap: coalesce author identities |
| Commit timestamp | July 29, 2026 (repository metadata) |
| Required vehicle repository | IGVC_ROS2 at d40e18aa480f76673b46a379ab7cb4618f279d75 (d40e18a), committed June 1, 2026 |
| Maintained package version | perception_costmap 0.1.0 |
| Documented offline result | 39 perception tests reported green in repository docs |
| Independent test run for this handoff | Not run: the local Windows documentation environment lacked pytest/ROS dependencies |
| Vehicle config | 3 ZED X cameras; HSV segmentation; front-camera YOLO; cones and white lines enabled; lidar disabled |
| Grid | x -4..16 m, y -10..10 m, 0.1 m resolution, base_link, 10 Hz requested publish rate |
| Nav2 state | Odom-relative rolling local/global costmaps; cloud bridge; disarmed by default |

## What appears complete in source

- Multi-camera BEV fusion and robot-centric grid construction.
- HSV and TwinLite segmentation interfaces, YOLO detection, cone detection, and white-line handling.
- Per-class danger zones for people, vehicles, cones, and generic obstacles.
- Active Nav2 bridge/configuration with an explicit safe command remap.
- Car deployment scripts, boot service, visualization, dashboard, and GMSL camera recovery procedures.
- Tools for IPM overlays, benchmarking, TensorRT export, shadow robustness, and route/goal validation.

## What is not proven by the repository alone

- That the physical robot is currently running commit 2684f87 or that the boot service is healthy today.
- That the current configuration sustains at least 8 Hz with all enabled perception stages.
- That cone/line geometry is within 0.1 m in the competition field.
- That Navigation2 avoids cones and people during repeatable supervised live runs.
- That lidar is ready for production fusion; it is explicitly disabled in the reviewed vehicle config.
- That the external sensor/actuator repositories and topic contracts still match this snapshot.

# 6. Environment and dependencies

| Component | Documented target |
|---|---|
| Vehicle compute | NVIDIA Jetson AGX Orin 64 GB ('dinosaur') |
| OS / middleware | JetPack 6.1 / L4T R36.4.x, Ubuntu 22.04, ROS 2 Humble |

| Component | Documented target |
|---|---|
| GPU stack | CUDA 12.6 and TensorRT 10.3 recorded in deployment notes |
| Cameras | 3x Stereolabs ZED X over GMSL: front, left, right |
| Lidar | Velodyne VLP-16 |
| Localization | Xsens MTi-680G IMU/GNSS plus external EKF/TF bring-up |
| Drive | Teensy 4.1 to dual REV SparkMAX over CAN; differential/skid-steer |
| Python ML | Jetson-specific torch 2.8.0, torchvision 0.23.0, numpy<2, ultralytics without replacing NVIDIA torch |
| ROS middleware | rmw_cyclonedds_cpp with the avros CycloneDDS configuration on the vehicle |

Jetson dependency trap
Do not install generic PyPI torch on the Jetson. The repository records that this can produce a CUDA-incompatible wheel. Use the Jetson AI Lab index for JetPack 6 / CUDA 12.6, keep torch/torchvision at the documented compatible versions, and pin numpy below 2 to protect the ROS 2 Humble cv_bridge chain.

### Required external components

- ROS 2 Humble, Navigation2, ZED SDK/wrappers, VLP-16 driver, Xsens/robot_state_publisher/EKF bring-up.
- The separate IGVC_ROS2 repository for sensor bring-up and vehicle integration.
- TwinLiteNetPlus source only if using the TwinLite backend.
- Target-built TensorRT engines for yolov8n.pt and cone_det.pt; optionally TwinLite once its exporter path is validated.
- web_video_server, tmux, and systemd support for the as-deployed operations scripts.

## 7. Build, launch, and operating runbook

### Portable checkout verification

```
git clone https://github.com/arassal/carla-nav2-avl.git
cd carla-nav2-avl/ros2_ws/src/perception_costmap
PYTHONPATH=. python3 -m pytest test -q
```

### Build only the maintained ROS package

```
cd carla-nav2-avl/ros2_ws
colcon build --packages-select perception_costmap
source install/setup.bash
```

### Prepare target-specific models

```
export AVL_MODELS_DIR="$(git rev-parse --show-toplevel)/models"
cd src/perception_costmap
python3 tools/export_trt.py --weights "$AVL_MODELS_DIR/yolov8n.pt"
python3 tools/export_trt.py --weights "$AVL_MODELS_DIR/cone_det.pt"
```

Vehicle launch path
Bring up external sensors/TF and EKF, start the three ZED wrappers sequentially, launch perception and visualization, then start Nav2 disarmed. Verify each gate before proceeding. The exact connection, service, recovery, disarmed-planning, and arming commands are consolidated in Appendix A; the field command card is Appendix B.

## 8. Configuration that matters

| Setting / contract | Reviewed value | Why it matters |
|---|---|---|
| segmentation_method | hsv | Only turn-key path from shipped artifacts; TwinLite requires external source/engine. |

| Setting / contract | Reviewed value | Why it matters |
|---|---|---|
| obstacle_method | yolo | Camera obstacle detection is learned-model based. |
| yolo_cameras | front | Only the front camera runs general YOLO in the current vehicle config. |
| use_cones / use_white_lines | true / true | Course-specific perception is enabled in configuration and code. |
| use_lidar | false | Raw VLP-16 does not currently contribute to the active perception/Nav2 obstacle path. |
| publish_rate | 10 Hz | Requested rate, not proof of achieved throughput. |
| grid | 20 x 20 m; 0.1 m/cell | Vehicle-centered perception region: x -4..16, y -10..10. |
| offroad_cost | 97 | Must remain synchronized with costmap_to_cloud obstacle threshold or road-keeping silently disappears. |
| unknown_cost | 25 | Blind areas are mildly penalized rather than left as ROS unknown. |
| image_stale_sec | 1.0 | Raised to prevent flicker from the observed front-camera rate and scheduling jitter. |
| Nav2 global frame | odom | No map/SLAM/GPS anchor is configured; navigation is odom-relative. |
| Nav2 safety gate | NAV2_ARMED != 1 | Default remaps commands to /nav2/cmd_vel so motors cannot receive them. |

### Camera mount values in active vehicle YAML

| Camera | Position (x, y, z m) | Pitch / yaw | Topic prefix |
|---|---|---|---|
| front | 0.6795, 0.000, 0.4476 | 17 / 0 deg | /zed_front/zed_node/... |
| left | 0.098, 0.286, 0.6126 | 18 / 94 deg | /zed_left/zed_node/... |
| right | 0.098, -0.286, 0.6126 | 20 / -86 deg | /zed_right/zed_node/... |

Configuration-change rule
Treat cost thresholds, frames, topic names, model paths, camera geometry, and arming behavior as interface contracts. Change them with a test or measured field observation, update the runbook, and verify from a fresh clone. Avoid hardcoded home paths outside deploy scripts.

# 9. Testing and acceptance

### Repository-reported tests

The project documentation reports 39 offline perception tests. Test files cover CARLA conversion, cheirality/orientation, course classes, detection, footprint geometry, segmentation, temporal filtering, and utilities. This handoff build did not execute them because the local Windows documentation environment did not have the Ubuntu/ROS Python test stack installed.

### Minimum takeover verification

| Level | Check | Pass condition |
|---|---|---|
| Offline | pytest in perception_costmap | All documented tests pass from a clean Ubuntu checkout. |
| Build | colcon build --packages-select perception_costmap | Clean build with no package-resolution errors. |
| Model | Export and benchmark YOLO/cone engines on target | Engines load without fallback; timing is recorded per stage. |
| Topics | Sensor and output topic rates | No stale streams; /perception/costmap sustained at least 8 Hz target. |
| Geometry | Asymmetric left/right fixtures plus tape-measured objects | No mirror errors; each camera places objects within 0.1 m / one cell target. |
| Perception | Person, cone, and white-line test | Lethal marking within 300 ms; person clears within 500 ms; course boundaries remain closed. |
| Planning | Plan-only obstacle corridor | Path avoids detected obstacle with required clearance; no motor command on /cmd_vel. |
| Motion | Supervised low-speed field test | E-stop proven; one actuator; speed caps respected; no boundary crossing/contact. |
| Reliability | Cold boot and three repeated runs | Hands-off service recovery and consistent completion without silent failure. |

### Evidence to archive with each release

- Exact Git commit, vehicle config diff, external bring-up commit, model hashes, and TensorRT build versions.
- pytest and colcon logs, topic-rate captures, per-stage benchmark output, and TF diagnostics.
- RViz/dashboard screenshots or a rosbag showing cone, person, and white-line behavior.
- Field-test date, operator, safety observer, E-stop check, weather/lighting, and failures observed.

# 10. Safety, troubleshooting, and recovery

> Motion safety
> Never set NAV2_ARMED=1 merely to see whether the stack works. First verify TF, odometry, perception, planning, costmap thresholds, command topic isolation, and E-stop function. Keep a human on the E-stop for every initial or changed configuration run.

The detailed symptom-to-response table and exact recovery commands are consolidated in Appendix A4. The non-negotiable rules are: start TF before ZED wrappers, recover cameras through clean_camera_restart.sh, require /perception/costmap_cloud before trusting Nav2, keep offroad_cost synchronized with the bridge threshold, verify model loading, and allow exactly one actuator process.

# 11. Risks, gaps, and documentation drift

| Priority | Risk / discrepancy | Handoff action |
|---|---|---|
| P0 | Physical system state is not knowable from the repository. | Confirm deployed commit, service status, sensor health, E-stop, and current operator notes before any motion. |
| P0 | Nav2 can drive the real actuator when NAV2_ARMED=1. | Require an explicit test authorization and human E-stop coverage; preserve disarmed default. |
| P1 | Active vehicle config disables lidar. | Decide whether camera-only avoidance is acceptable. If enabling lidar, add and validate the complete perception/Nav2 observation path. |

| Priority | Risk / discrepancy | Handoff action |
|---|---|---|
| P1 | Performance evidence is stale relative to current config. | Benchmark current HSV + front YOLO + cone/line stack on dinosaur and record sustained topic rate. |
| P1 | Calibration status conflicts across files. | Repeat tape-measure/asymmetric validation and update README/DESIGN/DEPLOY consistently. |
| P1 | FINALIZE.md says cones/white lines are not integrated, while source/config show integration. | Treat source as newer, test behavior, then retire or update stale phase text. |
| P1 | DESIGN.md describes StaticLayer, while active Nav2 uses costmap_to_cloud. | Update architecture documentation and remove ambiguity around deprecated nav2_costmap_params.yaml. |
| P1 | External avros/IGVC repository is a runtime prerequisite. | Pin and document the compatible commit and topic/TF contract. |
| P2 | Host-specific deploy scripts encode /home/dinosaur and network assumptions. | Keep them as the as-run path but add a parameterized installation procedure or environment file. |
| P2 | TwinLite is documented as faster-quality work but is not turn-key. | Either productize its exporter/runtime or explicitly keep HSV as the supported baseline. |
| P2 | Legacy/stub packages increase onboarding confusion. | Mark them clearly in package metadata or move them under a legacy area after team agreement. |

### Recommendations and decisions for the next team

- Supported production perception backend: HSV versus TwinLite/TensorRT.
- Camera-only versus camera-plus-lidar safety architecture.
- Canonical navigation architecture and deprecation plan for obsolete configs/tools.
- Field-calibration owner, procedure, acceptance tolerance, and recertification trigger.
- Release evidence, deployment owner, rollback method, and safety sign-off authority.

# 12. Handoff checklist

### Ownership and access

- [ ] Confirm project lead, perception owner, navigation owner, vehicle operator, and safety approver.
- [ ] Obtain repository, robot SSH/Tailscale, dashboard, and lab access through approved channels.
- [ ] Record the deployed commit for this repo and the external IGVC_ROS2/avros stack.
- [ ] Inventory robot services and confirm only one actuator node can run.

Technical and operational gates are not repeated here: use the acceptance matrix in section 9 and the executable checklists in Appendix A. Record every result against the exact deployed commits and model hashes.

### Suggested 30-day takeover plan

| Window | Outcome |
|---|---|
| Days 1-3 | Reproduce tests/build, map deployed versions, meet owners, and verify robot access without motion. |
| Week 1 | Recreate target engines, validate topics/TF/config, benchmark current perception, and resolve documentation drift. |
| Week 2 | Complete geometry and obstacle acceptance tests; decide the lidar and segmentation production baseline. |
| Weeks 3-4 | Run supervised field acceptance, harden boot/recovery, archive evidence, and publish an internally approved operating runbook. |

# A. Operator connection and startup guide

> Scope of this guide
> This procedure explains the operating path encoded in the two repositories. It does not confirm that the vehicle is currently powered, reachable, mechanically safe, or running the reviewed commits. Team authorization, credentials, a trained operator, and a tested physical E-stop are required before motion.

## Operator-computer software

| Tool | Purpose | Operator action |
|---|---|---|
| Tailscale | Private network path to the Jetson | Join the team-approved tailnet. Obtain access from the administrator; do not create a parallel network. |
| OpenSSH | Command-line control | Use Windows Terminal/PowerShell, macOS Terminal, or Linux shell. PuTTY is optional. |
| Web browser | Joystick/E-stop and perception dashboard | Use a current browser. Confirm the Jetson identity before accepting its self-signed joystick certificate. |
| NoMachine Client | Jetson desktop and RViz | Connect to the Jetson graphical session. RViz waits for this display and uses software rendering. |

## Repository-recorded connection

| Item | Recorded value | Caution |
|---|---|---|
| Tailscale address | 100.93.121.3 | Snapshot value; confirm in the tailnet before use. |
| Linux user | dinosaur | SSH credentials or keys must come from the team. |
| SSH alias | jetson | Only works if the operator's ~/.ssh/config defines it. |
| Vehicle workspace | /home/dinosaur/IGVC | The older /home/dinosaur/AVROS directory is documented as obsolete. |
| Perception workspace | /home/dinosaur/carla-nav2-avl | Deployment scripts assume this exact path. |

## Connect

```
# From the operator computer
tailscale ping 100.93.121.3
ssh dinosaur@100.93.121.3

# If the team-installed SSH alias exists
ssh jetson
```

> Connection failure
> If SSH fails, confirm Jetson power, Tailscale status, the current address, and SSH-key authorization. Do not guess passwords repeatedly or bypass the approved network.

# A1. Physical pre-start and boot

For the first startup after a handoff, place the vehicle on rated stands or use an approved clear test area. The operator must be able to stop both tracks immediately.

## Physical checklist

- [ ] Physical E-stop identified, reachable, and function-tested by the team operator.
- [ ] Motors disabled or tracks safely clear of the ground for software checkout.
- [ ] Battery condition and polarity verified; no damaged or loose power wiring.

- [ ] Jetson DC connector secured. Repository notes record previous hard power cuts from this connector.
- [ ] Teensy USB/serial cable secured; SparkMAX controllers and CAN wiring inspected.
- [ ] ZED Link/GMSL connections secured; camera lenses clean and unobstructed.
- [ ] Velodyne power and Ethernet path inspected. A July 16 note reported eno1 NO-CARRIER.
- [ ] Clear exclusion area established around the tracks; safety observer assigned.

### Expected boot services

| Service | Starts | Important ownership |
|---|---|---|
| avros-webui | actuator_node, webui_node, HTTPS port 8000 | Owns the Teensy serial interface and accepts web joystick or /cmd_vel. Exactly one actuator_node. |
| percept-stack | Sensors, EKF, three ZED wrappers, perception, visualization, ports 8080/8090, RViz waiter | Uses the dedicated tmux socket named percept. Three cameras start about 40 seconds apart. |

### Check service state

```
sudo systemctl status avros-webui --no-pager
sudo systemctl status percept-stack --no-pager
tmux -L percept list-windows -t percept
```

Expected tmux windows include sensors, ekf, zed_front, zed_left, zed_right, costmap, viz, wvs, www, costmap_rgb, and rviz. A complete camera startup may take several minutes.

## A2. Operator interfaces and live monitoring

| Interface | Address / command | Purpose |
|---|---|---|
| Perception dashboard | http://100.93.121.3:8090 | Fused BEV, rendered costmap, and three camera streams. |
| Joystick and software E-stop | https://100.93.121.3:8000 | Direct actuator control through avros_webui. Self-signed certificate is expected. |
| MJPEG backend | http://100.93.121.3:8080 | Stream server used by the dashboard; normally not operated directly. |
| RViz desktop | NoMachine -> 100.93.121.3 | Graphical ROS inspection. The checked-in launcher waits for a NoMachine display. |
| Process console | tmux -L percept attach -t percept | Inspect the active sensor, camera, perception, and visualization windows. |

### Useful tmux controls

| Keys | Effect |
|---|---|
| Ctrl-b, n | Next window |
| Ctrl-b, p | Previous window |
| Ctrl-b, w | List windows |
| Ctrl-b, d | Detach without stopping the stack |

### Log inspection

```
ls -l /tmp/sensors.log /tmp/ekf.log /tmp/zed_*.log /tmp/costmap.log
tail -n 100 /tmp/costmap.log
grep -Ei "engine|model|weights|fallback|error|warning" /tmp/costmap.log
```

Model-path check
perception_dinosaur.yaml uses AVL_MODELS_DIR, but the reviewed boot script does not visibly export it. Confirm that the service environment provides the variable and that /tmp/costmap.log reports successful model/engine loading rather than a silent fallback.

# A3. ROS environment and health gates

## Prepare every diagnostic shell

```
source /opt/ros/humble/setup.bash
source /home/dinosaur/IGVC/install/setup.bash
source /home/dinosaur/carla-nav2-avl/ros2_ws/install/setup.bash
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
export
CYCLONEDDS_URI=file:///home/dinosaur/IGVC/install/avros_bringup/share/avros_bringup/config/cyclonedds.xml
export AVL_MODELS_DIR=/home/dinosaur/carla-nav2-avl/models
```

## Required checks before Nav2

| Gate | Command | Required observation |
|---|---|---|
| Localization | ros2 topic hz /imu/data<br>ros2 topic hz /odometry/filtered | Stable rates; no unexplained gaps or stationary rotation. |
| TF | ros2 run tf2_ros tf2_echo odom base_link | Continuous, coherent transform. |
| Front/left/right RGB | ros2 topic hz /zed_front/zed_node/rgb/color/rect/image<br>ros2 topic hz /zed_left/zed_node/rgb/color/rect/image<br>ros2 topic hz /zed_right/zed_node/rgb/color/rect/image | All three publish current frames; topic existence alone is insufficient. |
| Perception | ros2 topic hz /perception/costmap | Sustained rate measured; project target is at least 8 Hz. |
| Perception geometry | Dashboard or RViz | Forward/left/right orientation correct; cone, person, and white line appear in the correct grid location. |
| Actuator | pgrep -af actuator_node<br>ros2 topic hz /avros/actuator_state | Exactly one actuator process; healthy state publishing. |
| Serial owner | sudo lsof /dev/serial/by-id/usb-Teensyduino_USB_Serial_18639150-if00 | Exactly one expected holder. |
| Lidar physical link | ip link show eno1<br>ethtool eno1 | Link detected: yes, if lidar is expected. |
| Lidar topic | ros2 topic hz /velodyne_points | Live points. Note that active perception config still sets use_lidar: false. |

## Visual acceptance

- A left-side obstacle must render left, and a right-side obstacle must render right.
- Cones and people must become lethal; painted white lines must close the course boundary.
- Removed obstacles must clear within the accepted time; stale frames must not remain trusted.
- Blind regions must not appear as confidently free because of a visualization or threshold error.

# A4. Manual restart and camera recovery

Stationary-only operation
The restart procedures stop and replace live perception processes. Run them only while the vehicle is stationary and autonomous motion is disabled.

### Full perception restart

```
bash /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap/deploy/full_stack_restart.sh
```

The script kills the existing percept tmux session, stops old ZED/perception/stream processes, starts sensors and EKF, waits, launches the three cameras sequentially, then starts perception and visualization. Manual mode places tmux in a systemd user scope so it survives SSH teardown.

### Verify completion

```
tmux -L percept list-windows -t percept
tail -n 100 /tmp/costmap.log
ros2 topic hz /perception/costmap
```

### Camera recovery

Use this for CAMERA STREAM FAILED TO START, Argus timeouts, blank video, or a stopped camera topic:

```
bash /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap/deploy/clean_camera_restart.sh
```

The recovery script stops all wrappers, restarts nvargus-daemon and zed_x_daemon, then starts the cameras about 40 seconds apart. Do not restart zed_x_daemon under live wrappers by hand; the repository records that this can wedge the remaining streams.

### Common symptom map

| Symptom | First check |
|---|---|
| ZED waits for static transforms | Confirm robot_state_publisher and the external sensor/TF stack started first. |
| Dashboard loads but images are blank | Check web_video_server, image topic rates, and the camera logs. |
| Perception grid is empty | Check current camera frames, model fallback messages, and config topic names. |
| Known grid flickers | Check camera rates/timestamps and image_stale_sec; current vehicle config uses 1.0 s. |
| Nav2 sees no obstacles | Confirm costmap_to_cloud is running and /perception/costmap_cloud publishes. |

# A5. Safe Navigation2 activation

### Start disarmed

```
cd /home/dinosaur/carla-nav2-avl/ros2_ws
unset NAV2_ARMED
ros2 launch src/perception_costmap/deploy/real_nav2_launch.py
```

The disarmed launch remaps controller output to /nav2/cmd_vel. The actuator listens to the real /cmd_vel, so this mode permits planning and controller inspection without intentional motor commands.

### Verify Nav2

```
ros2 lifecycle get /controller_server
ros2 lifecycle get /planner_server
ros2 lifecycle get /bt_navigator
ros2 topic hz /perception/costmap_cloud
ros2 topic info /nav2/cmd_vel -v
ros2 topic info /cmd_vel -v
```

All lifecycle nodes must be active. The perception cloud must publish. Use RViz to send a short odom-relative goal and confirm the planned path clears a test obstacle while Nav2 remains isolated from /cmd_vel.

### Pre-arm gate

- [ ] Approved clear area, trained operator, safety observer, and tested physical E-stop.
- [ ] Browser E-stop open and verified; only one actuator node and serial owner.
- [ ] Stable TF/odometry, three current camera streams, acceptable costmap rate, and correct geometry.

- [ ] Cone/person/line behavior verified in both perception and the Nav2 costmap.
- [ ] Disarmed plan avoids the obstacle; speed and goal distance approved.
- [ ] Lidar-disabled/camera-only limitation explicitly accepted for this test.
- [ ] Independent distance watchdog running in a separate shell.

### Distance watchdog example

```
cd /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap
python3 deploy/distance_watchdog.py --limit 2.0
```

# A6. Arming, first goal, E-stop, and shutdown

> Motion boundary
> The following commands can cause the physical tracks to move. They require the team's operating authority, a human on the physical E-stop, and a cleared test area.

## Arm Nav2

Stop the disarmed Nav2 launch with Ctrl-C. NAV2_ARMED is read when the launch file starts, so it must be set before relaunch:

```
cd /home/dinosaur/carla-nav2-avl/ros2_ws
export NAV2_ARMED=1
ros2 launch src/perception_costmap/deploy/real_nav2_launch.py
```

Verify /cmd_vel publisher/subscriber ownership immediately. Do not launch the separate avros_bringup navigation stack at the same time, and do not start a second actuator node.

## First short goal

The goal helper's default is about 6.1 m, which is inappropriate for a first handoff test. Pass a deliberately short approved distance, for example:

```
cd /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap
python3 deploy/send_goal_ahead.py 1.0
```

Watch the physical robot, RViz, browser E-stop, /cmd_vel, /odometry/filtered, watchdog, and actuator logs simultaneously.

## Emergency stop

- Primary: use the physical E-stop.
- Secondary: use the browser E-stop at https://100.93.121.3:8000.
- Command-line backup: publish a one-shot ActuatorCommand with estop: true.

```
ros2 topic pub --once /avros/actuator_command avros_msgs/msg/ActuatorCommand '{estop: true}'
```

## Normal shutdown

- Stop Nav2 with Ctrl-C and unset NAV2_ARMED.
- Confirm the robot is stopped and the motor system is disabled according to the team procedure.
- Stop perception with sudo systemctl stop percept-stack if ending the session.
- Stop joystick/actuator service with sudo systemctl stop avros-webui only after safe motor stop.
- Shut down Linux with sudo poweroff; wait for shutdown before removing power.

# B. Field command card

Use condition
This page is a reminder for a trained operator, not a substitute for Appendix A. Start on stands or in an approved clear area. Keep Nav2 disarmed until every gate passes.

All ROS commands below assume the shell preparation in Appendix A3 has already been completed.

| Task | Command / address |
|---|---|
| SSH | ssh dinosaur@100.93.121.3 |
| Service health | sudo systemctl status avros-webui percept-stack --no-pager |
| Perception console | tmux -L percept attach -t percept |
| Perception dashboard | http://100.93.121.3:8090 |
| Joystick / software E-stop | https://100.93.121.3:8000 |
| NoMachine / RViz | Connect to 100.93.121.3 as dinosaur |
| Perception log | tail -n 100 /tmp/costmap.log |
| Full perception restart | bash /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap/deploy/full_stack_restart.sh |
| Camera recovery | bash /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap/deploy/clean_camera_restart.sh |
| Costmap rate | ros2 topic hz /perception/costmap |
| Disarmed Nav2 | cd /home/dinosaur/carla-nav2-avl/ros2_ws; unset NAV2_ARMED; ros2 launch src/perception_costmap/deploy/real_nav2_launch.py |
| Distance watchdog | cd /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap; python3 deploy/distance_watchdog.py --limit 2.0 |
| Arm Nav2 | cd /home/dinosaur/carla-nav2-avl/ros2_ws; export NAV2_ARMED=1; ros2 launch src/perception_costmap/deploy/real_nav2_launch.py |
| Short goal | cd /home/dinosaur/carla-nav2-avl/ros2_ws/src/perception_costmap; python3 deploy/send_goal_ahead.py 1.0 |
| CLI E-stop backup | ros2 topic pub --once /avros/actuator_command avros_msgs/msg/ActuatorCommand '{estop: true}' |
| Return to safe default | Stop Nav2; unset NAV2_ARMED |
| Stop services | sudo systemctl stop percept-stack avros-webui |
| Power off Jetson | sudo poweroff |

## Do not proceed if

- No physical E-stop operator; more than one actuator node; serial-port collision.
- Missing/stale camera, unstable TF or odometry, incorrect left/right geometry, empty Nav2 obstacle cloud.
- Unexplained model fallback, costmap below accepted rate, lidar assumed active without evidence.
- The physical robot or deployed software differs materially from this reviewed snapshot.

# 13. Source references

This handoff was produced from read-only local clones of the two public GitHub repositories. No repository files, branches, issues, pull requests, or remote state were changed.

| Source | Location |
|---|---|
| Repository | https://github.com/arassal/carla-nav2-avl |
| Root orientation | https://github.com/arassal/carla-nav2-avl/blob/main/README.md |
| Reproduction path | https://github.com/arassal/carla-nav2-avl/blob/main/REPRODUCE.md |
| Architecture | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/DESIGN.md |
| Package guide | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/README.md |
| Jetson deployment | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/DEPLOY.md |
| Real vehicle config | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/config/perception_dinosaur.yaml |
| Active Nav2 config | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/config/real_nav2_params.yaml |
| Nav2 launch and safety gate | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/deploy/real_nav2_launch.py |
| Operations guide | https://github.com/arassal/carla-nav2-avl/blob/main/ros2_ws/src/perception_costmap/deploy/README.md |
| Contribution workflow | https://github.com/arassal/carla-nav2-avl/blob/main/CONTRIBUTION_GUIDE.md |
| Status/phase ledger | https://github.com/arassal/carla-nav2-avl/blob/main/FINALIZE.md |
| Required vehicle stack | https://github.com/Paarseus/IGVC_ROS2 |
| Vehicle operating notes | https://github.com/Paarseus/IGVC_ROS2/blob/main/CLAUDE.md |
| WebUI service | https://github.com/Paarseus/IGVC_ROS2/blob/main/scripts/systemd/avros-webui.service |
| Vehicle preflight | https://github.com/Paarseus/IGVC_ROS2/blob/main/scripts/preflight_check.sh |

> Review note
> Claims about completed behavior are phrased as repository evidence, not fresh vehicle verification. Where source and documentation conflict, this document identifies the conflict and recommends a confirming test.

## End of handoff